zudio

WESTSIDE

Customer Shopping Behaviour Analysis

TATA TRENT RETAIL ANALYTICS · MBA (AI&DS) MINI-PROJECT · SESSION 2025-2027

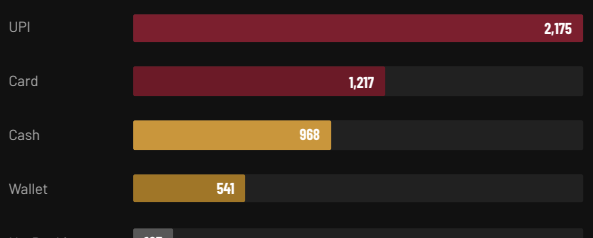
5,000	₹6.78M	86.1%	48.2%	₹1,360	4.15/5	20	18.25%
TRANSACTIONS	TOTAL REVENUE	MODEL ACCURACY	REPEAT PURCHASE RATE	AVG ORDER VALUE	AVG RATING	DATA COLUMNS	AVG DISCOUNT

01 · BRAND PERFORMANCE

3,147 ZUDIO ORDERS

1,853 WESTSIDE ORDERS

02 · PAYMENT METHODS



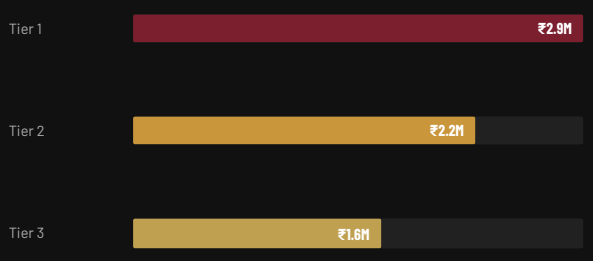
43.5% UPI DOMINANCE

03 · SALES CHANNEL

68.5% Offline
3,423 orders

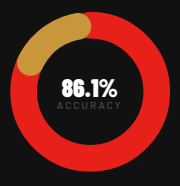
31.5% Online
1,577 orders · Higher AOV ↑

04 · REVENUE BY CITY TIER



05 · AVG SPEND BY AGE GROUP

06 · LOGISTIC REGRESSION MODEL

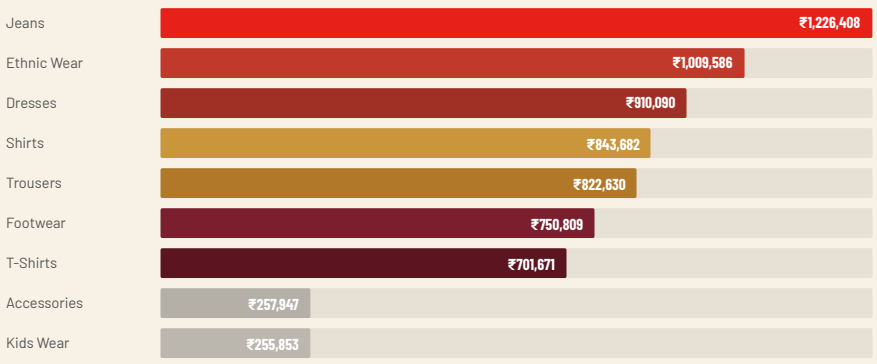


Precision	0.86
Recall	0.87
F1-Score	0.86

420	441	64	75
TRUE POS	TRUE NEG	FALSE NEG	FALSE POS

80/20 TRAIN-TEST SPLIT · 5,000 RECORDS · max_iter=1000

07 · PRODUCT CATEGORY REVENUE (INR)



08 · SEASONAL REVENUE TREND

FESTIVE	SUMMER	WINTER	MONSOON
₹2.21M	₹1.87M	₹1.43M	₹1.28M
32.6% of total	27.6% of total	21.1% of total	18.7% of total

DATA PIPELINE & ML PROCESS FLOW

DATA COLLECTION — 5,000 retail transactions · Zudio & Westside stores · 20 features per record — **5K** RECORDS

▼

DATA CLEANING — Null handling · Outlier removal · Type casting · Duplicate elimination — **100%** CLEAN

▼

EDA VISUALIZATION — Distribution analysis · Correlation heatmaps · Trend charts · Seaborn / Matplotlib — **20+** CHARTS

▼

FEATURE ENGINEERING — AOV · Customer Total Spend · Discount Amount · High-Value Flag · OneHotEncoding — **6** NEW FEATS

▼

MACHINE LEARNING MODEL — Logistic Regression · 80/20 split · 5,000 records · max_iter=1000 — **86.1%** ACCURACY

▼

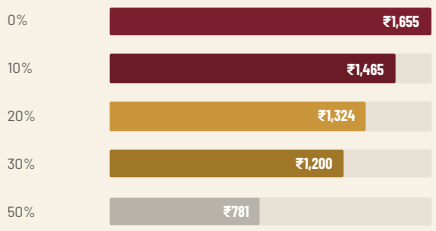
BUSINESS INSIGHTS — Loyalty drivers identified · Discount optimisation · Seasonal strategy · Power BI dashboards — **420** LOYAL CUST.

09 · MODEL EVALUATION

CONFUSION MATRIX

	Pred NO	Pred YES
Actual NO	441 True Neg	75 False Pos
Actual YES	64 False Neg	420 True Pos

DISCOUNT VS AVG SPEND



Sweet Spot: 10-20%

10 · REPEAT PURCHASE DRIVERS

- **High Total Spend** — Strongest predictor of repeat loyalty
- **High Product Rating** — Direct link to repeat behaviour
- **Moderate Discount Used** — Promotional sensitivity signal
- **Purchase Frequency** — Habitual buyer pattern indicator

11 · TOOLS & METHODOLOGY

Python · Pandas | Scikit-learn | Power BI | EDA | Feature Engineering | Logistic Regression | OneHotEncoder | SHAP / LIME | NumPy | Matplotlib · Seaborn | 80/20 Train-Test Split | Confusion Matrix

STRATEGIC INSIGHTS & BUSINESS IMPACT

Key Findings & Recommendations

ZUDIO & WESTSIDE · TATA TRENT LTD

12 · KEY INSIGHTS

Ratings Drive Loyalty
High-rated purchases correlated strongly with repeat behaviour. Improving in-store & online experience directly boosts retention.

Moderate Discounts Win
10-20% discounts outperform heavy offers. Extreme deals (40-50%) failed to boost revenue proportionally.

Metro Markets Lead
Tier-1 cities drive ₹2.9M. Premium expansion strategies in urban locations yield the highest ROI.

UPI Dominant (43.5%)
Digital-first payment is essential. Seamless UPI checkout significantly improves conversion rates.

Festive = Peak Sales
Festive generates ₹2.21M — 32.6% of total. Pre-season inventory & campaigns are mission-critical.

Jeans & Ethnic Lead
Top two categories total ₹2.24M. Depth stocking and fresh collections here maximise revenue capture.

13 · SCOPE & LIMITATIONS

Scope
- Customer transaction analysis
- Behavioural segmentation
- Dashboard visualisation
- Predictive modelling
- Multi-brand comparison
- Seasonal revenue analysis

Limitations
- Simulated retail dataset
- No long-term tracking
- External factors excluded
- Linear model constraints
- No competitor benchmarking
- Geography not granular

Assumptions
- Each transaction = unique customer visit
- Seasonal labels are self-reported in dataset
- UPI & digital payments reflect urban behaviour
- Discount rates applied uniformly per purchase
- Brand loyalty inferred from repeat purchase flag
- AOV is consistent across same city tiers

14 · FUTURE ROADMAP

Real-time Database
Auto-refresh BI dashboards

Recommendation Engine
Personalised product picks

Advanced ML Models
Random Forest / XGBoost

Cloud Analytics
Centralised BI platform

SUBMITTED BY

BHOOMI GUPTA

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